

Robust Redirected Walking in the Wild

Niall L. Williams*

University of Maryland, College Park

ABSTRACT

Locomotion is a fundamental component of experiences in virtual reality (VR). However, locomotion in VR is often difficult because the layouts of the physical and virtual environments are often different, which may cause unobstructed paths in the virtual world to correspond to obstructed paths in the physical world. Thus, in order to deliver a comfortable and immersive virtual experience to users, it is important that the user can explore the virtual world using techniques that help them avoid collisions with unseen physical objects. Redirected walking (RDW) is one such technique that enables collision-free locomotion in VR using real walking. Although RDW shows promise as an effective locomotion interface, it has seen relatively little adoption in the consumer market due to the difficulty in deploying effective RDW algorithms that are robust to different environment layouts and different users' perceptual thresholds. For my thesis, I am focused on developing RDW methods that are capable of enabling collision-free locomotion in arbitrary physical and virtual environments for a wide range of users.

Index Terms: Locomotion—Redirected walking—Natural walking—Motion planning;

1 INTRODUCTION

Redirected walking (RDW) [6] is a locomotion interface that allows users to explore large virtual environments (VEs) while located in smaller physical environments (PEs). RDW works by slowly rotating and translating the VE around the virtual camera while the user walks, which causes them to subconsciously adjust their physical trajectory in order to maintain their intended virtual path. When the user gets too close to a physical object, the virtual reality (VR) system initiates a *reset* maneuver, wherein the user turns around in place to face away from the nearby object before continuing to walk through the physical and virtual environments. These resets are undesirable since they tend to break the user's feeling of presence and make navigation more difficult. The faster the VE is rotated and translated, the easier it is to steer users away from objects, but the more likely it is that users perceive the VE movements and begin to feel sick. Thus, the goal of RDW is to rotate and translate the VE such that the user is steered along a physical path that minimizes the number of resets incurred, while minimizing the likelihood that the user consciously perceives the redirection.

RDW has received considerable attention within the academic community, with an abundance of research studying different steering algorithms and users' perceptual sensitivity to the visual gains induced by RDW [5]. Despite the numerous research papers about RDW that are published each year, RDW sees comparatively little use in consumer products. Among other factors, there are two major limitations of RDW that contribute to its slow adoption outside of academia. First, the RDW steering algorithms responsible for keeping users safe during locomotion need to be able to work in *any* type of physical and virtual environment. This is because RDW steering algorithms need to know the layout of the PE in order to steer

the user away from obstacles. Typically, steering algorithms only consider simple physical environments (e.g. an empty, rectangular room) which makes it difficult to use them in more compelling use-cases for VR, such as complex and dynamic environments where the layout is not known a priori (e.g. a user's apartment). Additionally, steering algorithms often ignore the layout of the VE completely, which further reduces their efficacy since the RDW problem depends on the *differences* between the PE and VE. For RDW to be usable at a consumer level, the steering algorithms need to be functional in arbitrary and unseen environments to account for the different settings in which users may use VR.

Second, RDW steering algorithms need to deliver adequate performance while accommodating for a diverse user population. During RDW, the virtual scene is rotated and translated around the user at speeds that are hopefully imperceptible. If these scene motions are too fast, users will consciously perceive them and will likely begin to feel simulator sickness or experience breaks in presence. To avoid this, the speed of virtual scene movements (controlled by parameters called *gains*) are constrained to stay within the user's estimated perceptual thresholds. Traditionally, estimation of a user's thresholds is a time-consuming and laborious process. Furthermore, research has shown that thresholds vary widely from person to person, as well as for the same person experiencing VR in different conditions (e.g. level of cognitive load or field of view in VR) [4, 13]. Thus, in order for RDW to be usable at scale in the consumer market, it is important that we develop methods for estimating users' RDW perceptual thresholds that are both *accurate* and *fast*.

For my PhD thesis, I am developing robust algorithms for RDW, to enable safe exploration of large VEs for a range of users in a variety of PEs. My thesis aims to improve the state-of-the-art in RDW by addressing two main limitations of RDW: (1) existing RDW algorithms cannot handle arbitrary, unknown, and unpredictable physical and virtual environments, and (2) existing methods for estimating RDW perceptual thresholds are time-consuming and yield inconsistent thresholds both with and between users.

2 METHODS FOR ROBUST REDIRECTED WALKING

2.1 Formalized Steering Algorithms Based on Motion Planning

Motion planning is a field in robotics that deals with computing a collision-free path through an environment to take a robot from a start configuration to a goal configuration [3]. Motion planning is a very well-studied topic, with a variety of path planning algorithms optimized for different robot and environment combinations. A large contributor to the success of robot motion planning algorithms is the rigorous and precise mathematical formalization upon which the planning problem is defined. This formalization allows researchers to take advantage of computational techniques like optimization and computational geometry, which enables the development of more robust and effective path planning algorithms.

As part of my PhD, I have formalized the RDW problem in terms of motion planning [11], providing a bridge between the RDW literature and the rich motion planning literature. Using this formalization, I have developed steering algorithms that take advantage of motion planning concepts and techniques to yield improved performance (fewer collisions) than existing algorithms [10, 11]. Additionally, I have used this formalization to develop a rigorous metric to estimate

*e-mail: niallw@cs.umd.edu

the complexity of navigation for a given pair of physical and virtual environments without requiring researchers to collect locomotion data, which can be time-consuming and expensive [12].

As I demonstrated in these publications [10–12], reframing RDW as a motion planning problem gives us new tools which we can use to advance the state of the art in VR locomotion. For the remainder of my PhD, I plan to further improve upon these steering algorithms and make use of additional concepts that are commonplace in modern motion planning. To this end, I plan to develop a RDW system that maps out the user's physical surroundings in real time using depth sensors, predicts the user's future trajectory, and steers the user in the PE to minimize the likelihood of collisions. In addition to incorporating modern motion planning techniques (real-time mapping and prediction), I plan to develop new RDW algorithms that make use of multiple different redirection techniques such as regular RDW [6], saccadic redirection [8], and flexible spaces [7, 9]. Using modern path planning techniques and multiple redirection methods, I hope to be able to enable users to explore large VEs using natural walking while located in arbitrary and dynamic PEs.

2.2 Efficient and Accurate Estimation of Perceptual Thresholds

In order for RDW to be effective, it is important that the gains which control the speed of rotations and translations of the VE are low enough that the user does not consciously perceive them. In order to maximize the steering potential of RDW while minimizing the likelihood that users are aware of the redirection, RDW steering algorithms are usually constrained to apply redirection using gains that lie within the user's estimated perceptual thresholds. These thresholds are measured using standard psychophysical methods, which involve fitting a curve to the participant's response data such that the curve models the participant's perceptual system [2].

Although the threshold estimation process is well-studied, there are two main problems which make it difficult to extend traditional estimation methods to the RDW setting. First, studies have shown that RDW thresholds vary widely depending on system and human factors such as exposure time, the head-mounted display's field of view, and the user's gender and cognitive load [1, 4, 13]. Given that a person's thresholds have a high degree of variability, it is necessary that we have accurate estimates of thresholds for the entire gamut of situations that users can be using VR in. However, the typical process of controlled experiments that measure thresholds under only a few specific conditions is not enough for researchers to be able to measure thresholds across all situations. Second, the estimation process usually requires participants to complete hundreds of trials in order for researchers to collect enough data to compute an accurate fit. Requiring users to complete hundreds of trials before they can use a locomotion interface is inconvenient and tiresome, and severely limits the usability of RDW.

Both of these problems call for the development of new methods for estimating RDW thresholds that are *accurate* and *fast*. I am currently studying the viability of new methods for computing psychophysical fits that make use of users' confidence ratings to estimate thresholds using five times fewer trials [14, 15]. In the future, I plan to investigate the efficacy of Bayesian methods for efficient threshold estimation. Additionally, I will explore the use of physiological signals (such as postural sway and eye movements) as a measure of a user's tolerance of redirection. Current threshold estimation methods all require users to answer questions about their perception, which is not representative of real-world scenarios and likely affects the measured thresholds since users are primed to specifically attend to any redirection sensations. Methods for measuring tolerance of RDW that are automatic and use objective physiological signals that users cannot control will be useful for continually measuring thresholds and actively adjusting the range of gains applied during locomotion. By being able to compute thresh-

olds quickly and accurately, we will be able to develop RDW systems that are capable of providing a comfortable locomotion experience for a wide range of users in a wide range of virtual experiences.

3 CONCLUSION

In this position paper, I outlined two major limitations of RDW that prevent it from being adopted on a large scale. First, most RDW steering algorithms are not functional in arbitrary, unseen, and dynamic physical and virtual environment pairs, which severely limits the different locations in which users can take advantage of RDW. Second, RDW relies on estimated perceptual thresholds which are difficult to measure accurately and quickly, since these thresholds vary widely depending on the particular user and various system factors. For my thesis, I am tackling the first problem by developing robust RDW steering algorithms that are based on rigorous concepts adapted from robot motion planning. Additionally, I am working on the second problem by investigating the efficacy of new statistical models and methods for quickly and accurately estimating a user's perceptual thresholds in real time. With this work, I aim to develop a locomotion system using RDW that enables collision-free navigation in arbitrary environments using real walking.

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